RESEARCH EXPERIMENT PLAN

When Is It Too Late?

Causal Recoverability Curves for Long-Horizon Vision-Language-Action Manipulation

A phased, falsifiable roadmap from the current SAFE and RoboCasa365 evidence to a competitive CoRL, RSS, ICRA, CVPR, ICCV, or WACV submission.

Primary target: CoRL | Natural alternative: ICRA | Stretch: RSS

CVPR/ICCV requires a central visual representation or flow-guidance contribution. WACV is a strong benchmark and evaluation fallback.

Prepared 23 August 2026 | Shinoda Lab RoboCasa research

Contents

This document is organized as a dependency-gated experiment program. Each phase specifies the scientific question, design, implementation controls, outcomes, continuation gate, failure response, and required artifacts.

Contents 2

Executive decision 5

The core distinction . . . 5

Recommended contribution package . . . 5

What should not be the headline . . . 5

Starting point and evidence status 6

Evidence chain required for the paper . . . 6

Roadmap overview 7

Phase 0. Freeze the research contract and audit readiness 8

Experimental design . . . 8

Implementation and validity controls . . . 8

Outcomes and analysis . . . 8

Predeclared continuation gate . . . 9

If the gate fails . . . 9

Required artifacts . . . 9

Phase 1. Frozen stage-aware detector confirmation 10

Experimental design . . . 10

Implementation and validity controls . . . 10

Outcomes and analysis . . . 10

Predeclared continuation gate . . . 11

If the gate fails . . . 11

Required artifacts . . . 11

Phase 2. Exact simulator-plus-policy snapshot replay 12

Experimental design . . . 12

Implementation and validity controls . . . 12

Outcomes and analysis . . . 12

Predeclared continuation gate . . . 13

If the gate fails . . . 13

Required artifacts . . . 13

Phase 3. Counterfactual recovery-opportunity screen 14

Experimental design . . . 14

Implementation and validity controls . . . 14

Outcomes and analysis . . . 14

Predeclared continuation gate . . . 15
If the gate fails . . . 15
Required artifacts . . . 15

Phase 4. Dense intervention-specific recoverability surfaces 16

Experimental design . . . 16
Implementation and validity controls . . . 16
Outcomes and analysis . . . 16
Predeclared continuation gate . . . 17
If the gate fails . . . 17
Required artifacts . . . 17

Phase 5. Learn a recovery-value critic 18

Experimental design . . . 18
Implementation and validity controls . . . 18
Outcomes and analysis . . . 18
Predeclared continuation gate . . . 19
If the gate fails . . . 19
Required artifacts . . . 19

Phase 6. Prospective best-of-K action-selection pilot 20

Experimental design . . . 20
Implementation and validity controls . . . 20
Outcomes and analysis . . . 20
Predeclared continuation gate . . . 21
If the gate fails . . . 21
Required artifacts . . . 21

Phase 7. Frozen end-to-end trigger-and-recovery evaluation 22

Experimental design . . . 22
Implementation and validity controls . . . 22
Outcomes and analysis . . . 22
Predeclared continuation gate . . . 23
If the gate fails . . . 23
Required artifacts . . . 23

Phase 8. Differentiable critic guidance inside the flow policy 24

Experimental design . . . 24
Implementation and validity controls . . . 24
Outcomes and analysis . . . 24
Predeclared continuation gate . . . 25
If the gate fails . . . 25
Required artifacts . . . 25

Phase 9. Generalization, second-policy replication, and physical validation 26
Experimental design . . . 26
Implementation and validity controls . . . 26
Outcomes and analysis . . . 26
Predeclared continuation gate . . . 27
If the gate fails . . . 27
Required artifacts . . . 27

Cross-cutting baseline matrix 28
One comparison per claim . . . 28

Statistical analysis plan 29
Estimands and units . . . 29
Power and sample-size strategy . . . 29
Multiplicity and analysis order . . . 29

Validity, leakage, and deployability checklist 30

Execution order and compute management 31
Can run in parallel . . . 31
Must remain sequential . . . 31
Canonical storage and run discipline . . . 31

Venue-specific completion bar 32

Proposed paper story and figures 33
Recommended title . . . 33
Three claims . . . 33
Minimum main-paper figures . . . 33
Suggested main-paper ablation table . . . 33

Immediate next-action checklist 34

References and repository anchors 35
Status note . . . 35

Executive decision

Recommended paper center

Do not center the paper on adapting SAFE or on detecting failure as early as possible. Center it on the causal value of intervention: whether to intervene, when each recovery operator remains useful, and which action or operator improves outcome from an identical long-horizon state.

The core distinction

A failure detector estimates whether the current trajectory will eventually fail. A recovery critic estimates whether a specific intervention now improves the probability of success relative to continuing. A SAFE score may be an input or baseline, but it is not automatically a recovery critic.

```
R_k(h_t,d) = P(success | do(k at t+d), h_t)
```

```
Delta_k(h_t,d) = R_k(h_t,d) - R_continue(h_t,d) - lambda * cost(k) - mu * harm(k)
```

Recommended contribution package

- An exact-state counterfactual recoverability benchmark with multiple intervention times and multiple recovery choices.
- Evidence that terminal failure likelihood, intervention value, and time remaining before recovery becomes ineffective are different quantities.
- A stage-aware utility gate that predicts Delta, controls false intervention, and routes to a recovery operator or action candidate.
- A frozen prospective trial showing higher final task success at matched intervention and compute budgets.
- Critic-guided flow as an optional executor, not the headline novelty unless it adds a clearly distinct algorithmic and matched-compute gain.

What should not be the headline

- Another retrospective SAFE ROC sweep.
- The statement that earlier recovery is always easier. Recoverability may be non-monotonic because policies self-correct and stages change.
- Environment rewind or oracle simulator stage as a deployable method.
- A terminal-failure score used as a flow critic without candidate-utility evidence.
- Gradient guidance before best-of-K critic selection beats random sampling prospectively.

Starting point and evidence status

The repository contains substantial detector and recovery infrastructure, but the evidence chain is not yet closed. The table distinguishes what has been measured from what remains a hypothesis.

Component	Current evidence	Scientific status	Consequence
Xiaomi frozen prospective SAFE	760 balanced primary rollouts; staged SAFE modestly improves matched-FPR classification over time-only, but the timing interval crosses zero and very few failures are found by the quarter-horizon landmark.	Valid shadow classification; failed overall early-warning gate; no intervention.	Do not claim deployable early recovery. Use as baseline and candidate feature.
Full-parent stage-aware detector	Development macro ROC rises from 0.592 terminal to 0.696 conditioned and wins on all five development tasks.	Selection evidence from 37 selection parents; not held-out or prospective.	Run the frozen 250-parent confirmation before using it as a trigger.
RLDX Subtask-SAFE	Retrospective raw ranking weakens under parent-aware causal analysis and is below the elapsed-time control; fixed-prefix evidence is near chance.	Negative causal detector evidence.	Do not mix RLDX and Xiaomi claims. Replicate only after a healthy policy positive control.
Recovery benchmark	Ordered subtask anchors and continue, EEF, and environment-rewind modes exist.	Code and local tests; historical live outcomes are limited and oracle-heavy.	Reuse the infrastructure, but separate deployable recovery from oracle upper bounds.
Xiaomi best-of-K	Deterministic K candidates and random, lowest-SAFE, and highest-SAFE selection are implemented with provenance records.	No prospective recovery result yet; current score is a within-state ranking probe.	Run exact branching and oracle-headroom screens before the 38-task scale-up.

Evidence chain required for the paper

Gate	Must establish	If it fails
1. Detector confirmation	The stage-aware signal transfers to new parents.	SAFE becomes a baseline/input, not the trigger contribution.
2. Replay validity	Same state and randomness reproduce; branches are attributable.	Stop and fix snapshot-policy state.
3. Oracle opportunity	Candidate/operator choice can change outcome.	Change proposals, timing, or operator.
4. Critic utility	Learned ranking beats random K under held-out branching.	Benchmark paper; no flow guidance.
5. Prospective intervention	Complete method improves full-task success without excessive harm.	No online-system claim.
6. External validity	Gain survives task/policy/robot shift.	Scope claim and venue accordingly.

Roadmap overview

Phase	Experiment	Planning scale	Main decision
0	Freeze and audit	One protocol and preflight	Is the study auditable?
1	Stage-aware confirmation	5 tasks x 50 parents = 250	Can SAFE be a trigger feature?
2	Snapshot replay	20 snapshots, about 120 branches	Are branches causal?
3	Opportunity screen	200 snapshots, about 1,000 branches	Is there critic headroom?
4	Recovery surfaces	12-16 tasks, about 3,000-5,000 suffixes	When and what recovery helps?
5	Recovery-value critic	Grouped development and sealed tests	Can Delta be predicted?
6	Best-of-K pilot	4 arms, 200 rollouts	Does critic selection work online?
7	End-to-end trial	12 tasks x 30-50 identities x main arms	Does final task success improve?
8	Flow guidance	2-task development, held-out transfer	Can K=1 guidance replace K=4?
9	Generalization and robot	Second policy and/or 90+ physical trials	Is the result externally valid?

Compute-saving rule

The first decisive runs are the 250-parent frozen confirmation and the bounded replay/opportunity screens. Do not spend the 38-task or flow-guidance budget until both exact replay and oracle recovery headroom are demonstrated.

Phase 0. Freeze the research contract and audit readiness

Purpose: prevent promising pilot evidence from silently defining the final test.

Freeze the scientific question, estimands, task allocations, model artifacts, thresholds, policy interface, and evidence labels before new collection. This phase turns the current repository into an auditable starting point rather than another mutable development run.

Field	Frozen plan
Scientific question	Can every future result be traced to an immutable policy, feature, task, seed, intervention, and analysis definition?
Primary hypothesis	Not applicable. This is a reproducibility and validity gate.
Entry condition	Current Xiaomi SAFE runtime, five-task stage-aware development bundle, recovery benchmark, and best-of-K implementation exist locally or on durable cluster storage.
Exit condition	A signed-off protocol and machine-readable allocation manifest exist; server, storage, checkpoint, and one-rollout smoke checks pass.

Experimental design

Component	Specification
Primary policy	Xiaomi-Robotics-1 SAFE-enabled RoboCasa365 checkpoint. RLDX is reserved for later replication.
Development tasks	ArrangeBreadBasket, ArrangeTea, BreadSelection, CuttingToolSelection, and GarnishPancake.
Primary paper endpoint	Absolute full-task success improvement of the complete recovery-value system over the strongest frozen non-oracle baseline.
Primary causal object	Intervention advantage Delta_k at an exact prefix, relative to continuing from the same state.
Experimental hierarchy	Task -> parent identity -> snapshot identity -> candidate/intervention branch.
Evidence labels	Development selection, retrospective locked, prospective shadow, prospective intervention, simulator oracle, and real-robot evidence must remain separate.

Implementation and validity controls

- Freeze repository commit, Xiaomi checkpoint hash, SAFE commit, runtime bundle, task/stage catalog, feature layer, replan_steps, seed families, reset indices, ports, output roots, and code patches.
- Create disjoint manifests for development, calibration, confirmation, closed-loop test, unseen-task test, and physical evaluation. Identity is the tuple (task, environment seed, reset index, policy-sampling family).
- Follow docs/cluster_experiment_runbook.md: source code stays under /gs/fs; rollouts, videos, logs, checkpoints, and reports stay under /gs/bs.
- Require model-loaded logs, listening ports, GPU processes, storage bytes and inodes, and a bounded one-rollout smoke before every new collection block.
- Disable W&B; unless explicitly needed. Preserve partial manifests and errors; never replace a failed run in place.

Outcomes and analysis

- A protocol.json containing hypotheses, primary comparisons, sample units, metrics, gates, and analysis order.
- An allocation manifest containing every task, arm, environment seed, reset index, policy seed, candidate seed, server block, and planned output path.
- A provenance record containing code/checkpoint hashes and the exact distinction between deployable and oracle information.

Predeclared continuation gate

All frozen artifacts resolve and their checksums are recorded.

No identity overlaps exist across development, calibration, confirmation, and final test allocations.

One bounded rollout per server completes with zero traceback and produces aligned action, feature, stage, and result records.

The primary endpoint and one primary method comparison are fixed before new outcomes are inspected.

If the gate fails

Failure is a result, not permission to retune the test set

Do not collect at scale. Repair missing provenance, overlapping identities, server instability, storage problems, or feature/action misalignment first. Infrastructure failures are not experimental failures and must not enter scientific denominators unless the attrition rule says so.

Required artifacts

- protocol.json and protocol.md
- allocation_manifest.jsonl
- artifact_checksums.json
- preflight and smoke logs under /gs/bs/tga-shinoda/felid/robocasa_logs/eval

Phase 1. Frozen stage-aware detector confirmation

Decision: does the selected composite-task signal survive truly new identities?

Confirm the existing stage-aware development result without changing the runtime. The outcome determines whether SAFE is allowed to serve as a deployable trigger feature; recovery-value research may continue even if this detector gate fails.

Field	Frozen plan
Scientific question	Does full-parent stage-aware conditioning improve exact-prefix ranking over terminal SAFE and stage-conditioned time on new parents?
Primary hypothesis	At prefixes 1 and 2, the frozen conditioned detector has task/stage-macro ROC at least 0.60 and improves over terminal SAFE by at least 0.05.
Entry condition	Phase 0 complete; frozen stage-aware runtime from development; new seed/reset family reserved.
Exit condition	The 250-parent confirmation is scored once, with parent/task bootstrap intervals and no test-set update.

Experimental design

Component	Specification
Tasks	The same five Xiaomi composite tasks used in development.
Collection	50 consecutive, identity-disjoint parents per task; 250 total. Retain every valid attempt rather than stopping by outcome quota.
Primary prefixes	Exact active-stage policy-inference prefixes 1 and 2. Prefixes 4 and 8 are secondary risk sets.
Frozen methods	Conditioned stage-aware SAFE, terminal SAFE, broad SAFE38, prototype distance, and task/stage-conditioned elapsed time.
Context	Record causal observation/state history for later development, but forbid it in this frozen confirmation.
Unit and uncertainty	Parent rollout is the sampling unit. Bootstrap task first and parent within task; do not count inference rows as independent.

Implementation and validity controls

- Use the evaluate phase in docs/safe_xiaomi_stage_aware_parent.md with the frozen runtime bundle. The evaluator must reject development/test identity overlap.
- Do not refit checkpoints, task normalizers, time curves, stage support, prototypes, thresholds, or score transformations on confirmation parents.
- Preserve all five tasks, including CuttingToolSelection and ArrangeBreadBasket as mandatory weak-case diagnostics.
- Distinguish the consecutive confirmation stream from any later case-control dataset; this collection can describe natural attempted-parent rates within its fixed collection protocol.
- Report supported and unsupported stages explicitly. Later-prefix analyses must display the shrinking at-risk denominator.

Outcomes and analysis

- Primary: mean task/stage-macro ROC across exact prefixes 1 and 2.
- Primary contrast: conditioned minus terminal macro ROC, with hierarchical 95% interval.
- Secondary: conditioned versus time-only, SAFE38, per-task and per-stage ROC/AP, calibration plots, and 10/25/50% stage recall.
- Descriptive only: matched-FPR event metrics, because the development calibration pool is too small for a precise 5% operating point.

Predeclared continuation gate

Conditioned task/stage-macro ROC >= 0.60.

Conditioned minus terminal ROC >= +0.05, and the paired hierarchical interval excludes zero.

Conditioned beats terminal on at least four of five tasks.

Conditioned beats time-only on at least four of five tasks.

If the gate fails

Failure is a result, not permission to retune the test set

Do not continue architecture search on these confirmation identities. Remove stage-aware SAFE from the deployable trigger claim, retain terminal SAFE and stage-aware SAFE as negative baselines, and continue the causal recoverability program with time, progress, observation, and action-context features.

Required artifacts

- immutable raw parent collection and semantic traces
- frozen score files for every comparator
- parent/task bootstrap replicates and per-stage support table
- one-page gate report with PASS or FAIL and no post-hoc exceptions

Phase 2. Exact simulator-plus-policy snapshot replay

Decision: can alternative recoveries be attributed to the intervention rather than hidden state drift?

Extend state restoration from simulator state to the complete causal execution state, then prove that identical branches reproduce and deliberately different candidate seeds generate aligned alternatives.

Field	Frozen plan
Scientific question	Can the same parent prefix be replayed with common random numbers so branch differences identify intervention effects?
Primary hypothesis	Identical snapshot and seed bundles reproduce the next observation, action, progress, and suffix outcome within predefined tolerances.
Entry condition	Phase 0 complete; current environment capture/reset and Xiaomi deterministic candidate-seed paths are available.
Exit condition	A bounded two-task reproducibility test passes and a versioned full-snapshot schema is frozen.

Experimental design

Component	Specification
Tasks	ArrangeTea and CuttingToolSelection, chosen to include a relatively promising and a difficult composite.
Parents	Five new identities per task.
Snapshots	Two per parent: active-stage prefix 2 and a fixed 25%-stage landmark. Total 20 snapshots.
Branches	Two repeated nominal continuations plus four explicitly seeded candidate chunks per snapshot. Approximately 120 branch executions.
Comparison	Repeated nominal branches use identical common randomness. Candidate branches differ only in the declared candidate seed at the first recovery inference.
Tolerance	Exact checksums for requests/actions when deterministic; numeric tolerance for simulator arrays; exact ordered-stage trace and success labels.

Implementation and validity controls

- Capture environment state, MuJoCo state, environment RNG, simulator RNG, policy image/state queues, cached action plan, instruction, inference counters, candidate index, stage-tracker state, and all client/server sampling identifiers.
- Restore the full state before every branch. Recompute observations only through one frozen path and verify that reset does not mutate predicates or completed-stage order.
- Store the snapshot schema version and checksums. Never use Python object identity or undocumented server state as the only reproducibility guarantee.
- Record one SAFE feature per genuine policy inference, not per cached replan action.
- Separate restore fidelity from rendering fidelity; video problems must not invalidate otherwise correct simulator measurements.

Outcomes and analysis

- Next-action and next-observation agreement for repeated branches.
- qpos/qvel and observable-state deviation after one step and after the suffix horizon.
- Ordered stage/progress agreement and full suffix success agreement.
- Candidate action diversity, feature diversity, sampling-seed alignment, server errors, and branch runtime.

Predeclared continuation gate

Repeated same-seed branches reproduce next observation/progress in at least 95% of snapshots and show no systematic task-specific failure.

Candidate action chunks are non-identical on at least 90% of snapshots.

Every candidate seed maps to exactly one recorded feature tensor, action chunk, and branch outcome.

Restoration alone never regresses a previously completed ordered stage.

If the gate fails

Failure is a result, not permission to retune the test set

Stop all counterfactual scaling. Diagnose policy cache, RNG, observation refresh, instruction, or server-state mismatch. A recovery paper cannot make causal claims from environment-only reset if the policy state or seed schedule differs between branches.

Required artifacts

- versioned full_snapshot schema and serializer
- snapshot_restore_validation.json
- per-snapshot checksum and tolerance report
- short paired videos for human inspection, labeled as visualization rather than numerical proof

Phase 3. Counterfactual recovery-opportunity screen

Decision: does candidate choice create enough outcome variation for a critic to learn?

Measure the oracle opportunity available from the current Xiaomi stochastic proposal distribution before investing in a recovery critic or flow guidance.

Field	Frozen plan
Scientific question	From the same state, do seeded action chunks produce meaningfully different recovery outcomes?
Primary hypothesis	A nontrivial subset of snapshots has mixed candidate outcomes, and oracle-best-of-four materially exceeds random candidate selection.
Entry condition	Phase 2 passes; candidate generation and continuation seed schedules are frozen.
Exit condition	A 1,000-branch screen quantifies oracle headroom, diversity, success-control harm, and task coverage.

Experimental design

Component	Specification
Parents	For each of five tasks, 10 eventually failed and 10 successful-control parents. Total 100. This is an explicit case-control opportunity dataset, not a prevalence sample.
Snapshots	Two per parent: exact active-stage prefix 2 and a fixed 25%-stage landmark. Total 200.
Candidates	Nominal K=1/default plus four explicitly seeded stochastic candidates from the same observation and instruction.
Execution	Execute the first chunk, then continue with one frozen fallback-policy seed schedule. Use identical recovery horizon and compute across candidate branches.
Primary label	Active-stage completion within 120 environment steps, or a horizon frozen from training-task duration statistics.
Secondary labels	Full-task success where affordable, ordered progress delta, prior-stage regression, collision/safety flags, latency, and policy calls.

Implementation and validity controls

- Use candidate seeds shared across scoring strategies. Save decoded actions, action latents/SAFE features, score components, task/stage context, and every suffix outcome.
- Weight each snapshot equally regardless of how many valid candidate records it produces. Preserve server errors and candidate ties.
- Include successful-control snapshots to quantify how intervention can break a trajectory that would otherwise succeed.
- Do not select snapshot times with the SAFE detector being evaluated; use the prespecified prefix/landmark grid.
- Run task shards on separate servers/ports/output roots only after one bounded smoke per server.

Outcomes and analysis

- Mixed-outcome fraction: snapshots containing both success and failure candidates.
- Oracle-best-of-four, random-of-four, nominal K=1, lowest-SAFE, and highest-SAFE expected recovery rates computed from identical candidate sets.
- Candidate diversity and SAFE score spread; correlation between terminal SAFE score and actual candidate advantage.
- Successful-control harm and completed-stage regression.

Predeclared continuation gate

At least 20% of snapshots exhibit mixed candidate outcomes.

Mixed outcomes occur in at least three of five tasks.

Oracle-best-of-four improves recovery by at least 15 percentage points over random candidate choice.

Candidate diversity remains high enough that score ties do not explain the result.

If the gate fails

Failure is a result, not permission to retune the test set

Do not train a critic. First change proposal diversity, snapshot timing, recovery prompt/operator, replan length, or continuation horizon, then repeat a new bounded screen. If oracle selection still has no headroom, abandon candidate reranking and focus on higher-level rollback or replanning operators.

Required artifacts

- counterfactual_opportunity_manifest.jsonl
- paired candidate/action/feature files
- oracle_headroom_by_task_stage.csv
- diversity, harm, regression, and failure-mode audit

Phase 4. Dense intervention-specific recoverability surfaces

Scientific core: characterize when each intervention remains useful.

Construct the paper's distinctive dataset: multiple recovery decisions and multiple intervention times branched from identical long-horizon states. Estimate recovery value and last-useful-intervention time without assuming monotonic decay.

Field	Frozen plan
Scientific question	How do recovery probability and intervention advantage vary with task, stage, delay, failure mode, and recovery operator?
Primary hypothesis	Recoverability is heterogeneous and sometimes non-monotonic; terminal failure probability does not reliably identify the best operator or its deadline.
Entry condition	Phase 3 shows meaningful oracle headroom; snapshot schema and intervention definitions are frozen.
Exit condition	A task-diverse, parent-grouped causal dataset supports both descriptive recovery curves and critic training.

Experimental design

Component	Specification
Task scope	12-16 composite RoboCasa365 tasks selected before method results, stratified by horizon, stage count, baseline success, and failure support.
Planning scale	Approximately 20 parent identities per task, three supported stage prefixes per parent, and four deployable branches: roughly 3,000-5,000 suffix executions after attrition.
State grid	Prespecified genuine policy-inference prefixes such as 1, 2, 4, and 8 within the active stage. Report the at-risk denominator and censoring at every prefix.
Recovery operators	Continue, ordinary replan, atomic/subtask re-prompt or rollback, random best-of-K, and later critic-selected best-of-K. Environment rewind is an oracle only.
Latency grid	For trigger-based branches, add delays d in {0, 1, 2, 4} genuine policy inferences rather than cached action steps.
Failure sources	Natural failures are primary. Injected actuation, visual, or state perturbations are a separate stress-test stratum and are never pooled with natural deployment rates.

Implementation and validity controls

- Define $R_k(h_t,d) = P(\text{success} \mid \text{do}(k \text{ at } t+d), h_t)$ and Δ_k as R_k minus continue probability and a frozen intervention-cost penalty.
- Use common random numbers for environment and fallback-policy sampling across branches. Record where common randomness ends after trajectories diverge.
- Predeclare the operational deadline as the last sampled delay where the lower-confidence estimate of Δ_k remains positive, plus an interval reflecting grid resolution.
- Separate observable/predicted stage context from simulator-oracle predicates. Oracle-stage curves are upper bounds; deployable claims require the causal stage tracker.
- Version every branch state, intervention, horizon, and outcome. Keep task, parent, snapshot, and candidate identifiers explicit in all tables.

Outcomes and analysis

- Recovery probability and causal advantage curves by operator, stage, task, and delay.
- Integrated recovery value over the supported delay grid and intervention-specific deadline error.
- Rescue and harm decomposition on failed and successful-control parents.

- Self-correction and non-monotonicity frequency; heterogeneity by failure family.
- Agreement between terminal SAFE ranking, elapsed-time ranking, and causal intervention-value ranking.

Predeclared continuation gate

At least three deployable intervention operators show measurable positive value somewhere in the task/stage grid, or the operator set is narrowed transparently.

At least four held-out task-stage strata contain sufficient mixed branch outcomes for evaluation.

Continue branches reproduce expected natural suffix behavior and successful-control harm remains measurable rather than silently filtered.

The dataset supports task/parent/snapshot-grouped train, validation, and untouched test partitions without identity leakage.

If the gate fails

Failure is a result, not permission to retune the test set

If useful recovery is confined to one task or one simulator-oracle mode, do not claim a general recovery surface. Narrow the paper to a benchmark/diagnostic result, improve deployable recovery operators, or change the primary policy to one with more recoverable natural failures.

Required artifacts

- recoverability_surface_manifest.jsonl and schema
- branch outcomes with common-random-number provenance
- task-stage-delay support and censoring tables
- recovery curves, deadline estimates, rescue/harm plots, and failure taxonomy

Phase 5. Learn a recovery-value critic

Method: predict causal utility, not eventual failure.

Train a critic that estimates stage completion, full-task success, and paired intervention advantage for candidate action chunks or recovery modes. The critic must distinguish candidates from the same state and calibrate uncertainty on held-out parents and tasks.

Field	Frozen plan
Scientific question	Can observable history and candidate representations predict which intervention will improve outcome from this exact state?
Primary hypothesis	A context- and candidate-conditioned critic outperforms terminal SAFE, time/stage heuristics, and action-only alternatives in within-snapshot ranking and top-1 recovery.
Entry condition	Phase 4 produces a leakage-safe surface dataset with sufficient mixed outcomes.
Exit condition	One frozen critic and uncertainty rule pass held-out snapshot and task-level gates.

Experimental design

Component	Specification
Inputs	Candidate action token/latent, decoded action statistics, causal observation/state history, task language, predicted stage, stage progress, and elapsed inference count.
Targets	Stage completion, full-task success, progress delta, prior-stage regression/safety penalty, and Delta relative to default/continue from the same snapshot.
Objectives	Snapshot-weighted BCE or value regression plus within-snapshot pairwise ranking; optional conservative penalty for critic uncertainty.
Splits	Split by parent before extracting snapshots. Keep every branch from a snapshot together. Reserve complete tasks/scenes for generalization.
Development order	Logistic/linear probes -> one small MLP per representation -> three seeds only for the top two frozen representations.
Selection	One primary validation score combining within-snapshot ranking and calibration. Do not select from final task-success results.

Implementation and validity controls

- Preserve terminal SAFE as a scalar baseline and raw SAFE/action features as a separate input ablation. Do not call the existing failure detector a recovery critic.
- Train with equal snapshot and task weighting so tasks with more candidate branches do not dominate.
- Fit task offsets, feature normalizers, progress curves, and uncertainty calibration on development/calibration partitions only.
- Build a deployable stage estimator and measure its stage confusion and latency separately. Report oracle-stage critic performance only as an upper bound.
- Use full-task outcomes on the held-out set even if stage completion is used as the cheaper training target; quantify proxy mismatch.

Outcomes and analysis

- Within-snapshot pairwise ROC and ranking accuracy.
- Top-1 recovery success, oracle regret, calibration error, Brier score, and expected Delta by predicted-value bin.
- Performance on held-out parents, held-out layouts/scenes, and held-out tasks.
- Ablations: time/stage only; terminal SAFE scalar; action latent only; observation/state only; action plus context; full model; oracle versus predicted stage.

Predeclared continuation gate

Held-out within-snapshot pairwise ROC >= 0.65 and lower interval above 0.50.

Learned top-1 improves recovery by at least 10 percentage points over random K=4, with a paired interval excluding zero.

The improvement is positive on at least four of five development tasks and remains positive on held-out task identities.

Successful-control harm does not increase by more than five percentage points relative to the matched random selector.

If the gate fails

Failure is a result, not permission to retune the test set

Do not proceed to differentiable flow guidance. Diagnose label horizon, proxy mismatch, proposal diversity, stage leakage, or insufficient counterfactual support. A robust benchmark paper remains possible if the failure-probability-versus-utility mismatch is strong and well supported.

Required artifacts

- frozen critic checkpoint and runtime bundle
- model card with input provenance and deployability constraints
- grouped split manifests and calibration bundle
- ablation, calibration, regret, and per-task evaluation tables

Phase 6. Prospective best-of-K action-selection pilot

Mechanistic test: isolate critic quality from trigger quality and from the benefit of sampling more candidates.

Run a prospective randomized selection study under a fixed trigger. This is the first closed-loop test that the critic selects actions with greater recovery utility rather than merely fitting retrospective labels.

Field	Frozen plan
Scientific question	Given identical candidate sets, does the learned critic improve recovery over random selection and the existing SAFE ranking?
Primary hypothesis	Critic-selected K=4 improves stage and task recovery over random K=4 under identical candidate seeds and compute.
Entry condition	Phase 5 critic, candidate count, recovery horizon, and trigger state are frozen.
Exit condition	A 200-rollout pilot supports or rejects prospective actionability before end-to-end scaling.

Experimental design

Component	Specification
Tasks and identities	Five development tasks, ten new parent identities per task.
Arms	K=1 nominal; K=4 random; K=4 lowest existing SAFE; K=4 highest learned recovery value.
Trigger	Fixed stage prefix or simulator-oracle trigger used equally across arms. Trigger quality is intentionally held constant.
Pairing	Same task/environment identity, ordinary-inference seeds, and first candidate set across arms. After action selection, analyze diverged suffixes by intention to treat.
Primary comparison	Learned critic K=4 versus random K=4.
Negative control	Highest terminal-failure-risk candidate, or shuffled critic scores, if candidate support permits.

Implementation and validity controls

- Interleave arms within task and server block so allocation age, GPU state, policy-server drift, and simulator changes cannot align with treatment.
- Keep candidate count, candidate seeds, recovery horizon, fallback seed schedule, and number of policy calls identical for the two K=4 primary arms.
- Record first-decision candidate outcomes separately from later closed-loop outcomes because candidate sets diverge after different actions are executed.
- Report candidate ties and identical actions. Do not impute a successful choice when the candidate set offered no successful branch.

Outcomes and analysis

- Primary: absolute stage-recovery difference, critic K=4 minus random K=4.
- Secondary: full-task success, oracle regret, progress, completed-stage regression, collision/safety flags, runtime, and policy calls.
- Mechanistic: probability that critic chooses a better candidate than terminal SAFE; score-outcome monotonicity; performance conditional on oracle headroom.

Predeclared continuation gate

Critic K=4 beats random K=4 with a paired hierarchical interval excluding zero.

A practically meaningful improvement of at least 10 percentage points is observed in the opportunity-supported subset.

No material increase in successful-control harm, regression, or safety events.

The effect is not driven entirely by one task or one server block.

If the gate fails

Failure is a result, not permission to retune the test set

Do not build flow guidance. If offline ranking passed but prospective reranking fails, treat this as distribution shift or proxy mismatch, preserve the negative result, and decide whether one new development cohort is justified. Never tune on the prospective identities.

Required artifacts

- interleaved arm allocation and intention-to-treat results
- candidate-set equality and divergence audit
- paired recovery and task-success tables
- videos for representative rescue, harm, tie, and critic-error cases

Phase 7. Frozen end-to-end trigger-and-recovery evaluation

Main paper result: does the complete system improve final task success at controlled intervention cost?

Evaluate the recovery-value gate and critic-selected executor prospectively on a fixed natural stream. This experiment links detection, timing, operator selection, and embodied outcome in one auditable trial.

Field	Frozen plan
Scientific question	Does a utility-aware trigger/router improve long-horizon task success over no intervention and the strongest calibrated trigger baseline?
Primary hypothesis	The complete method produces a positive, task-consistent absolute task-success lift while controlling successful-parent harm and intervention budget.
Entry condition	Phases 1-6 are frozen; the critic passes actionability; a fresh seed/task allocation and calibration cohort are reserved.
Exit condition	One locked analysis establishes the primary result and all rescue/harm, cost, timing, and failure-mode diagnostics.

Experimental design

Component	Specification
Task scope	At least 12 composite tasks, including current development tasks and at least four held-out task or scene identities.
Planning sample	30-50 unique parent identities per task and arm as a planning floor. Final N is frozen by blinded hierarchical power simulation from pilot variance.
Deployable arms	No intervention; invocation-matched periodic; stage/time trigger; terminal or stage-aware SAFE trigger; proposed recovery-value trigger plus critic-selected recovery.
Oracle arm	Best available intervention timing/operator from counterfactual branches. Report separately and never include in deployable averages.
Primary comparison	Complete method versus the strongest non-oracle baseline selected and frozen on development data.
Randomization	Blocked and interleaved by task, environment identity, server session, and GPU allocation; analyze by intention to treat after divergence.

Implementation and validity controls

- Freeze feature extractor, detector, critic, stage tracker, task/stage catalog, thresholds, intervention budget, K, guidance/recovery horizon, and cost penalty before collection.
- Use one common executor for periodic, time, and SAFE trigger arms so the trigger comparison is interpretable. The separate Phase 6 pilot identifies selector value.
- Define successful-parent FPR at the episode level: any unnecessary intervention during a would-be successful parent counts once.
- Use at least 200 successful calibration parents for a 5% operating point; aim for 400 if resources allow. Calibration and final evaluation identities must be disjoint.
- Use a fixed-attempt natural stream. Outcome-quota cohorts may support ranking but cannot estimate deployment prevalence or operational intervention rates.

Outcomes and analysis

- Primary: absolute full-task success difference, complete method minus strongest frozen non-oracle baseline.
- Rescue and harm: rescued failed parents; corrupted would-be successes; prior-stage regression; repeated-intervention outcomes.

- Timing: lead before intervention-specific deadline, miss-adjusted detection time, recall by 10/25/50% stage duration, and imposed-latency sensitivity.
- Decision quality: intervention-mode regret and calibration of predicted Delta.
- Cost and safety: number of alarms, candidates, policy calls, environment steps, wall-clock latency, collisions, saturation, and errors.

Predeclared continuation gate

Lower 95% hierarchical confidence bound on the primary task-success lift is above zero.

The absolute gain is practically meaningful, with 5 percentage points as the proposed minimum target.

The gain has the same sign in most task blocks and is not explained by one task, seed family, or server session.

Successful-parent harm and episode-level false intervention remain within the frozen budget.

If the gate fails

Failure is a result, not permission to retune the test set

Do not claim an effective online recovery system. If causal curves and offline critic remain strong, reposition as an evaluation/benchmark paper. If task success improves only under oracle reset or privileged stage identity, report the deployability gap as a limitation rather than averaging oracle and deployable arms.

Required artifacts

- locked trial allocation and immutable outcome manifest
- primary analysis with hierarchical bootstrap replicates
- rescue/harm, timing, calibration, cost, and safety tables
- representative success, rescue, harm, late-trigger, and no-headroom videos

Phase 8. Differentiable critic guidance inside the flow policy

Optional algorithmic extension: attempt only after the critic proves prospective actionability.

Convert the recovery-value critic into guidance over intermediate noisy action/flow states and test whether one guided sample can match or exceed best-of-K recovery at lower inference cost.

Field	Frozen plan
Scientific question	Can recovery-value gradients improve executed actions without candidate enumeration, distribution collapse, or safety regressions?
Primary hypothesis	A frozen noise-step-conditioned critic guides a single flow trajectory to recovery success comparable to critic-selected K=4 with less wall-clock inference.
Entry condition	Phase 6 passes prospectively; the Xiaomi denoising/flow internals expose a differentiable action latent and frozen sampling protocol.
Exit condition	A guidance scale is tuned only on development tasks and evaluated once on held-out tasks at matched compute reporting.

Experimental design

Component	Specification
Development tasks	ArrangeTea and CuttingToolSelection, 20 fresh identities per task.
Test tasks	The remaining current tasks plus held-out composite tasks from Phase 7.
Arms	Unguided K=1; random K=4; critic-reranked K=4; critic-guided K=1; shuffled/random-gradient negative control.
Critic state	Condition on observation/history, stage/progress, noisy action latent, and flow/noise integration step.
Guidance parameters	Scale, start/end integration steps, gradient norm clip, suffix scope, and action-range guardrail. Tune once on development identities.
Compute matching	Report number of function evaluations, backward passes, policy calls, GPU time, and wall-clock latency; provide matched-NFE and matched-time views.

Implementation and validity controls

- Move critic execution inside the policy model's denoising/flow loop. The current detached client-side SAFE scoring path is insufficient for gradient guidance.
- Train or distill the critic on intermediate noisy action states; final-action-only supervision can be used but its generalization across flow steps must be verified.
- Freeze gradient normalization, clipping, action bounds, and guidance scale before held-out testing.
- Record unguided and guided latent trajectories for analysis, but evaluate claims on executed environment outcomes rather than critic score improvement alone.

Outcomes and analysis

- Stage recovery and full-task success at matched NFE and matched wall-clock budgets.
- Change in predicted Delta versus realized Delta, action saturation, collision rate, prior-stage regression, and policy-distribution shift.
- Ablations over guidance scale, guided flow-step window, suffix length, SAFE risk target versus recovery-value target, and shuffled-gradient control.

Predeclared continuation gate

Guided K=1 matches or exceeds critic-reranked K=4 recovery within the frozen statistical margin.

Guidance reduces meaningful inference cost or latency rather than only candidate count.

No material increase in saturation, collision, oscillation, or completed-stage regression.

The frozen guidance setting transfers from two development tasks to held-out tasks.

If the gate fails

Failure is a result, not permission to retune the test set

Retain critic reranking as the recovery executor. Do not present critic-guided flow as a contribution if it improves critic scores but not task outcomes, or if matched-compute baselines erase the gain. The central causal recoverability paper does not require this phase to succeed.

Required artifacts

- model-side guidance implementation and tests
- noise-step critic checkpoint and frozen guidance configuration
- matched-compute and safety analysis
- latent trajectory diagnostics linked to executed rollouts

Phase 9. Generalization, second-policy replication, and physical validation

Venue-strengthening phase: show that the result is not a five-task Xiaomi artifact.

Test held-out tasks and scenes, replicate the causal result on another VLA family, and validate a small number of controlled long-horizon recoveries on a physical robot where feasible.

Field	Frozen plan
Scientific question	Do recoverability curves and the utility gate transfer across task identity, layout, policy representation, and embodiment?
Primary hypothesis	The recovery-value formulation transfers more reliably than raw SAFE thresholds, even if feature extractors and calibration remain policy-specific.
Entry condition	Phase 7 passes on Xiaomi; all external-validation protocols are frozen before outcomes.
Exit condition	At least one strong external-validity axis passes: second policy, substantial held-out task/scene shift, or controlled physical trials; strongest submissions include two.

Experimental design

Component	Specification
Held-out simulation	At least four unseen composite task identities plus layout/style shifts, language paraphrases, camera perturbations, and object distractors where scientifically relevant.
Second VLA	RLDX-1 is the natural replication candidate only after an ordinary no-SAFE positive-control evaluation proves healthy success and progress. Keep its features/checkpoints/calibration separate from Xiaomi.
Physical tasks	Three or four multi-stage tabletop tasks with no intervention, strongest baseline, and full method as the main conditions.
Trial planning	Aim for 15-20 trials per physical task and main condition, or at least about 90 controlled recovery trials total if hardware time is limited.
Transfer options	Zero-shot critic transfer, feature-adapter transfer, and policy-specific critic retraining must be reported as distinct settings.
Primary external endpoint	Task-macro success lift of the complete method on identities not used for model, threshold, task-stage catalog, or guidance selection.

Implementation and validity controls

- Run a no-SAFE policy health check and bounded smoke before any second-policy data collection. Stop abnormal zero-success or no-progress blocks rather than collecting unusable failures.
- On real hardware, define safe stop, human takeover, collision, object-drop, timeout, and reset procedures before trials. Log interventions and exclusions.
- Measure stage-estimation performance without privileged simulator predicates. Manual annotations may be used for evaluation but not online decisions.
- Preserve separate policy-family results. Never average incompatible feature semantics or task distributions into one headline number.

Outcomes and analysis

- Task-macro recovery and final success on unseen tasks/scenes and the second policy.
- Transfer degradation in critic calibration, trigger timing, operator selection, and deadline prediction.
- Physical rescue, harm, latency, intervention count, human takeover, and safety outcomes.
- Qualitative failure cases showing perception shift, stage-tracker error, critic error, and unrecoverable states.

Predeclared continuation gate

The primary success lift remains positive on the selected external-validity axis with task/trial-level uncertainty.

The deployable stage tracker, not oracle stage, supports the result.

Physical or second-policy gains are not obtained by increasing intervention count or compute beyond the matched baseline budget.

Safety/harm outcomes remain within the predeclared bound.

If the gate fails

Failure is a result, not permission to retune the test set

Scope the claim to Xiaomi RoboCasa365 and choose the venue accordingly. A failed transfer is informative if the paper identifies which component - feature, calibration, stage tracker, proposal, or critic - caused the gap. Do not conceal policy-specific calibration behind an aggregate score.

Required artifacts

- external-validation protocol and frozen identities
- second-policy positive-control and replication reports
- physical trial log, safety protocol, videos, and exclusion ledger
- transfer and domain-shift ablation tables

Cross-cutting baseline matrix

Use the fewest baselines needed for each causal question, but ensure the final paper covers detector quality, action selection, intervention timing, and recovery execution separately.

Layer	Required baselines	What the comparison isolates
Detector/ranking	Stage-conditioned elapsed time; terminal SAFE; segmented Subtask-SAFE; full-parent stage-aware SAFE; broad SAFE38; observation/state-only; action-disagreement; success prototype.	Whether SAFE representation adds causal early ranking beyond task progress and time.
Stage information	Oracle simulator stage; predicted causal stage; no stage; task only.	Deployability and privileged-information gap.
Candidate selection	K=1; random K=4; lowest terminal SAFE; highest terminal SAFE; learned critic; oracle best-of-K.	Sampling benefit, SAFE actionability, critic value, and available headroom.
Trigger	No intervention; invocation-matched periodic; stage/time hazard; SAFE; recovery-value gate; oracle deadline.	Whether timing policy improves success at the same intervention budget.
Recovery operator	Continue; ordinary replan; subtask retry/rollback; critic candidate; environment rewind oracle.	Which deployable mechanism helps at each state and delay.
Flow executor	Unguided K=1; random K=4; critic K=4; guided K=1; shuffled/random-gradient negative control.	Whether guidance improves actual outcomes at matched compute.

One comparison per claim

Claim	Primary evidence	Primary comparison
Stage-aware signal generalizes	Phase 1 frozen confirmation	Conditioned versus terminal at exact prefixes 1/2
Failure risk differs from recovery value	Phase 4 exact-state branches	SAFE ranking versus realized Delta
Critic selects actionable chunks	Phases 5-6	Critic K=4 versus random K=4
Utility gate improves execution	Phase 7 prospective trial	Complete method versus strongest frozen non-oracle baseline
Guidance is efficient	Phase 8 matched-compute trial	Guided K=1 versus critic K=4
Method generalizes	Phase 9	Full method versus strongest baseline on external identities

Statistical analysis plan

The experimental unit is not an inference row

Tasks, parents, snapshots, and candidate branches are nested. Treating thousands of inference records as independent would produce misleadingly narrow intervals. Candidate branches from one snapshot share state and must remain grouped.

Estimands and units

Estimand	Unit	Recommended analysis
Detector ranking	Parent with task/stage risk sets	Task-first, parent-within-task bootstrap; macro task/stage ROC and AP.
Candidate advantage	Snapshot with paired branches	Within-snapshot contrasts; snapshot -> parent -> task bootstrap.
Closed-loop success lift	Parent identity assigned to an arm	Blocked intention-to-treat difference; task and server-session hierarchy.
Successful-parent harm/FPR	Would-be successful parent	Episode-level any-intervention event; binomial and task-bootstrap intervals.
Deadline and timing	Failed parent-stage with supported recovery branches	Miss-adjusted time; survival/risk-set reporting; grid-resolution interval.
Generalization	Held-out task/scene/policy	Task-macro effect; report each group and random-effects heterogeneity.

Power and sample-size strategy

- Use fixed small samples for screening gates; do not interpret them as definitive publication estimates.
- For the final closed-loop trial, use a blinded internal pilot to estimate baseline rates, task/session variance, and paired discordance without selecting the treatment effect. Freeze final N through simulation of the exact hierarchical analysis.
- A planning floor is 30-50 identities per task across at least 12 tasks. If the expected absolute success lift is only 5-8 points, several hundred parents per main arm may be necessary.
- The previous Xiaomi timing result remained imprecise with 380 failed parents. If timing remains a primary claim, budget roughly 400-600 independent failures, then confirm by bootstrap power simulation.
- For a 5% successful-parent false-intervention threshold, 200 calibration successes are a practical minimum and 400 are preferable. Report the finite-sample upper confidence bound, not only empirical FPR.

Multiplicity and analysis order

- Declare one primary endpoint and one primary comparison: full-task success, complete method versus strongest frozen non-oracle baseline.
- Use hierarchical testing: primary outcome first; rescue/harm and timing second; per-task, ablation, and failure-type analyses third.
- For multiple secondary model contrasts, use Holm adjustment or label them exploratory. Neural initialization seeds are model uncertainty, not independent experimental replications.
- Publish all predeclared task and stage strata, including zero-support, error, and failed-estimation rows.

Validity, leakage, and deployability checklist

Risk	Required control
Parent leakage	Split parents before extracting stages, prefixes, snapshots, or candidates. Keep all descendants of one parent in one partition.
Snapshot leakage	All K candidates and operator branches from one snapshot stay together. Weight snapshots, not candidate rows.
Future information	Exclude final rollout length, future predicates, future stage completion, termination reason, and future features from deployed inputs.
Oracle stage leakage	Measure predicted stage separately. Oracle simulator predicates are upper-bound analyses only.
Survivor bias	Later prefixes include only still-active stages. Publish risk-set denominators and use survival/IPCW sensitivity if timing claims depend on later prefixes.
Outcome quota	Case-control collections support ranking/opportunity, not prevalence, event rate, or operational FPR.
Feature duplication	Store one feature per genuine inference. Cached replan actions do not create new monitor observations.
Server drift	Interleave arms within server blocks; record allocation, port, GPU, server log, model-loaded timestamp, and error status.
Restore confounding	Restore simulator, policy, queues, counters, instruction, and RNG. Prove same-seed reproducibility before branching.
Compute confounding	Match K, recovery horizon, policy calls, NFE, or wall-clock budget according to the declared comparison; report both calls and time.
Selective attrition	Use a prespecified error/partial-run rule. Preserve logs and partial manifests; never silently rerun only failed arms.
Oracle/deployable mixing	Never average environment rewind, ground-truth stage, or retrospective oracle timing with deployable results.

Execution order and compute management

The program is designed to stop early when the causal signal or engineering validity is absent. Parallelism is useful only where it does not violate dependencies or share a stateful server unexpectedly.

Can run in parallel

- Phase 1 confirmation collection and Phase 2 full-snapshot implementation/validation.
- Stage-tracker development and CPU-side schema/analysis tooling, while confirmation collection remains frozen.
- Task-sharded branch collection on separate GPUs, Xiaomi servers, ports, log files, and output roots after the one-server smoke passes.
- Critic training on completed development shards while validation/test parent identities remain sealed.
- RLDX ordinary-policy positive control on a separate allocation after Xiaomi method development is stable.
- Video indexing, bootstrap analysis, provenance checks, and failure annotation on CPU.

Must remain sequential

Dependency	Reason
Replay validity before branch scale	Otherwise treatment effects are confounded by hidden simulator or policy state.
Oracle opportunity before critic training	A critic cannot recover headroom absent from the candidate/operator set.
Held-out critic gate before prospective reranking	Prevents online identities from becoming critic-selection data.
Prospective reranking before flow guidance	Proves the critic's causal actionability before adding a differentiable mechanism.
Freeze before every new confirmation	No model, normalization, threshold, task, or seed choice may depend on test outcomes.

Canonical storage and run discipline

- Source repository: /gs/fs/tga-shinoda/felid/robocasa.
- Durable rollouts: /gs/bs/tga-shinoda/felid/robocasa_rollouts.
- Durable logs: /gs/bs/tga-shinoda/felid/robocasa_logs/eval.
- Failure/recovery datasets: /gs/bs/tga-shinoda/felid/robocasa_failure_datasets.
- Checkpoints and frozen runtime bundles: /gs/bs/tga-shinoda/felid/robocasa_checkpoints.
- Before every expensive job: verify storage margin, model-loaded log, listening port, server/GPU process, exact task list, unique output root, and bounded smoke.

Venue-specific completion bar

Venue	Competitive completion package	Risk if submitted too early
CoRL	Causal benchmark; learned utility gate; broad held-out task/scene evaluation; prospective closed-loop gain; preferably second VLA or physical evidence.	A five-task detector paper or oracle-heavy recovery system will look incremental beside CheckVLA and B2FF.
RSS	Exceptionally clean exact-state causal design, operational irreversibility/deadline analysis, latency and safety evidence, strong system story, preferably physical demonstration.	Broad but shallow benchmark tables will not substitute for a crisp robotics mechanism.
ICRA	Complete integrated system, broad RoboCasa365 study, runtime/safety analysis, strong component ablations, and several physical multi-stage tasks if possible.	Simulator-only one-backbone results may be judged as incomplete system validation.
CVPR / ICCV	Make visual learning central: counterfactual video/feature dataset, transferable visual recoverability representation, multiple encoders, and held-out camera/object/scene shifts. Closed-loop control validates the vision method.	A primarily trigger-and-control paper is likely a scope mismatch even if robotically strong.
WACV	Strong reproducible benchmark/application: exact-state protocol, public code/data, rigorous negative audit, critic reranking, and useful held-out evaluation. Flow or hardware is helpful but not essential.	Without released artifacts or a significant vision/application component, the contribution may seem narrow.

Recommended submission strategy

Build the complete experiment as a CoRL paper. If the causal benchmark and critic are strong but end-to-end or hardware evidence is incomplete, redirect to WACV evaluation/application. If the integrated system succeeds with a smaller physical study, ICRA is natural. Aim at RSS only with exceptional causal clarity and physical evidence. Aim at CVPR/ICCV only after the representation or flow method becomes central and transferable.

Proposed paper story and figures

Recommended title

When Is It Too Late?

Causal Recoverability Curves for Long-Horizon Vision-Language-Action Manipulation

Three claims

- Terminal failure likelihood is not equivalent to intervention value or remaining recovery time.
- Recoverability varies with stage, failure mode, delay, and operator, and need not decay monotonically.
- A learned stage-aware utility gate improves prospective task success while controlling unnecessary and harmful interventions.

Minimum main-paper figures

Figure	Content	Experiment source
1. Concept	Failure-risk monitor versus causal recovery-value gate and operator router.	System overview
2. Evidence audit	Retrospective, causal-prefix, prospective shadow, and intervention evidence separated.	Current report plus Phase 1
3. Recovery surfaces	Delta by stage, delay, and operator with self-correction/non-monotonic examples.	Phase 4
4. Risk versus value	SAFE failure probability versus realized intervention advantage and deadline error.	Phases 4-5
5. Critic selection	Random, SAFE, critic, and oracle top-1 recovery/regret.	Phases 5-6
6. Closed-loop result	Task-macro final success, rescue, harm, and intervention cost.	Phase 7
7. Generalization	Held-out tasks/scenes, second policy, or physical tasks.	Phase 9

Suggested main-paper ablation table

Rows: time/stage only; SAFE scalar; action latent; observation/state; action plus context; full recovery-value critic; oracle stage. Columns: pairwise ranking, top-1 recovery, task success, rescue, harm, deadline error, latency, and model calls. Keep full per-task tables in the appendix.

Immediate next-action checklist

Order	Action	Completion evidence
1	Freeze the Phase 0 protocol, artifact hashes, and identity allocations.	No-overlap report and signed-off primary comparison.
2	Collect 250 new stage-aware confirmation parents while recording context but not using it.	Immutable collection with five-task coverage and zero unreported errors.
3	Implement complete policy-plus-simulator snapshot state.	Versioned schema and checksum tests.
4	Run the 120-branch replay smoke.	At least 95% reproducibility and at least 90% candidate diversity.
5	Score the frozen detector confirmation once.	PASS/FAIL gate report; no refit.
6	Run the 1,000-branch opportunity screen.	Mixed-outcome and oracle-headroom report.
7	Freeze the larger task/operator/delay allocation only if oracle headroom passes.	Surface protocol and sealed test identities.
8	Train simple probes, then one small critic; delay broad sweeps.	Held-out ranking, calibration, and top-1 recovery.
9	Run the 200-rollout prospective best-of-K pilot.	Critic versus random K=4 intention-to-treat result.
10	Power and launch the final closed-loop trial; attempt flow guidance only afterward.	Frozen N, final allocation, and one locked analysis.

First irreversible research decision

Do not launch the current 38-task best-of-K screen or implement gradient guidance first. The next scientific value comes from the 250-parent frozen confirmation and the exact-state oracle-opportunity screen. Together they determine whether SAFE is a usable trigger and whether candidate recovery is learnable.

References and repository anchors

ID	Source	Location
[R1]	Full SAFE RoboCasa experimental report	reports/safe_robocasa_full_report_20260804/build/main.pdf
[R2]	Frozen stage-aware Xiaomi protocol	docs/safe_xiaomi_stage_aware_parent.md
[R3]	Xiaomi best-of-K recovery protocol	docs/xr1_safe_best_of_k_recovery.md
[R4]	Cluster experiment runbook	docs/cluster_experiment_runbook.md
[R5]	Recovery rollout and state-reset implementation	robocasa/recovery/recovery_rollout.py
[P1]	SAFE: official NeurIPS 2025 abstract	papers.nips.cc/.../SAFE
[P2]	CheckVLA	arxiv.org/html/2607.26789
[P3]	B2FF	arxiv.org/html/2606.09258
[P4]	CycleVLA	arxiv.org/html/2601.02295
[P5]	AFIL	arxiv.org/html/2605.08434
[P6]	RoVer	arxiv.org/html/2510.10975
[P7]	Q-VGM	arxiv.org/html/2606.08015
[V1]	CoRL call for papers	corl.org/contributions/call-for-papers
[V2]	RSS call for papers	roboticsconference.org/information/cfp
[V3]	CVPR 2026 scope	cvpr.thecvf.com/Conferences/2026/CallForPapers
[V4]	WACV 2027 call for papers	wacv.thecvf.com/Conferences/2027/CallForPapers

Status note

Sample sizes and gates labeled as proposed are planning decisions rather than official venue requirements. Final powered sample sizes must be frozen from blinded pilot variance and the exact hierarchical analysis. Current simulator, cluster, and prospective claims remain bounded by the evidence labels stated in this document.

Document completion criterion

The plan is complete when every phase has a frozen protocol, an auditable artifact bundle, a PASS or FAIL gate, and a scientifically honest next branch. A failed phase narrows the claim; it does not authorize retuning on its test identities.